

Module 2– Week 2

Exploratory Data Analysis Report

Dataset: Online Gaming Behaviour

Date: 2025-05-31

1. Introduction

1.1 Purpose and Source of the Dataset

The Online Gaming Behaviour dataset (40,034 rows, 13 columns) captures demographic information and gaming-related metrics for a large sample of players. Each record corresponds to a unique player and includes variables such as age, gender, geographic location, daily playtime in hours, weekly session count, in-game purchase behaviour, and indicators of engagement. This dataset was originally collected by an independent game analytics firm to understand player demographics and in-game behaviour across different genres. Because it has never been widely analysed in published academic literature, it satisfies the assignment requirement of using a “fresh” dataset.

Source: <https://www.kaggle.com/code/sulaniishara/player-engagement-analysis-prediction/notebook>

1.2 Reason for Choosing this Dataset

I selected this dataset because it:

- Exceeds 6,000 rows (in fact, 40,034), granting statistical power to detect patterns.
- Contains a mixture of numerical, categorical, and binary fields, enabling exploration of continuous, discrete, and categorical data types.
- Offers rich context (age, gender, genre, session frequency, playtime, player level, achievements, purchases, engagement), which allows for a wide variety of analyses (e.g., demographic patterns, purchase habits, engagement segmentation).

1.3 Objectives and Learning Goals

- **Gain familiarity** with raw, uncleaned player data.
- **Produce descriptive statistics** (means, medians, standard deviations) for key variables (e.g., PlayTimeHours, SessionsPerWeek, PlayerLevel).
- **Visualize distributions** (histograms, boxplots) to identify central tendencies, spread, and outliers.

- **Investigate relationships** (scatterplots with regression fits) between variables such as “Sessions per Week” and “Daily PlayTime.”
- **Clean the dataset** to remove non-active players (zero sessions, zero hours) and compare “before vs. after” insights.
- **Segment and compare** subgroups (by genre, age group, engagement level), computing subset statistics and plotting accordingly.
- **Draw actionable insights** about player behavior, such as which genres drive higher playtime or purchase rates.

2. Data Description & Cleaning

2.1 Overview of Data Fields

The original dataset contains 40,034 rows and the following **13 columns**:

Field	Type	Description
PlayerID	Integer	Unique identifier for each player (1–40,034).
Age	Integer	Player’s age (range: 15–49).
Gender	Categorical	“Male” or “Female.”
Location	Categorical	Geographic region: “USA”, “Europe”, “Asia”, or “Other.”
GameGenre	Categorical	Primary genre: “Sports”, “Action”, “Strategy”, “Simulation”, or “RPG.”
PlayTimeHours	Numeric	Daily average hours spent playing (0.0001 – 23.9996).
InGamePurchases	Binary (0/1)	1 = player makes in-game purchases; 0 = no purchases.
GameDifficulty	Categorical	Player’s self-reported difficulty setting: “Easy”, “Medium”, or “Hard.”
SessionsPerWeek	Numeric	Number of gameplay sessions per week (0 – 19).
AvgSessionDurationMinutes	Numeric	Mean length (minutes) of each session (10 – 179).
PlayerLevel	Numeric	Current in-game level (1 – 99).

Field	Type	Description
AchievementsUnlocked	Numeric	Total achievements unlocked (0 – 49).
EngagementLevel	Categorical	Composite label: “Low”, “Medium”, or “High.” Derived externally based on playtime and sessions.

2.2 Missing Values & Initial Integrity Checks

- **Missingness:** Each of the 40,034 rows has a non-missing value in all 13 fields. No imputation was required.
- **Uniqueness:** PlayerID is unique for all rows; there are no duplicate entries.

2.3 Data Cleaning Steps

1. Missing Values:

- A check for missing data revealed zero missing values across all 13 fields.
- Conclusion: No imputation or row-removal was necessary.

2. Data Types and Consistency:

- All columns have the expected types (e.g., categorical fields are stored as strings, numeric fields as floats/integers).
- The PlayerID field is unique (no duplicates), so no de-duplication was required.

3. Outlier Checks:

- Boxplots (shown below) confirmed that all numeric variables lie within expected ranges (e.g., Age ranges 15–49; PlayTimeHours ranges from ~0 to 24).
- No extreme “impossible” values (like negative ages or playtime), so no further filtering was applied.

Because the dataset was already complete and free of invalid values, no additional data cleaning steps were required.

During initial exploration, it became evident that a fraction of players have zero sessions per week or zero daily playtime—likely representing newly registered or inactive accounts. Including these “zero-activity” players can obscure patterns among genuinely active gamers. Thus:

1. Remove all rows where

- SessionsPerWeek < 1 **OR**
- PlayTimeHours < 1

After filtering, the cleaned data (df_clean) contains 32,235 players (approximately 80.5% of the original sample).

2. **Create an AgeGroup variable** by binning Age into four categories:

- 15-24, 25-34, 35-44, 45-49.

This allows direct comparison of playtime by life stage.

After cleaning, the dataset retains the same 13 original fields (plus the new AgeGroup), with 32,235 rows of bona fide “active” players. Subsequent analyses and visualizations use df_clean unless explicitly noted as an “all-players” plot.

3. Descriptive Statistics

3.1 Numeric Variables (Cleaned Dataset)

Below are summary statistics for the main numeric variables—calculated on df_clean (n = 32,235):

Variable	Count	Mean	Std Dev	Min	25th %	Median (50%)	75th %	Max
Age	32,235	32.2879	9.9732	15	23	32	41	49
PlayTimeHours	32,235	12.2024	6.7410	1.0001	6.3838	12.1495	17.9544	23.9996
InGamePurchases	32,235	0.2336	0.4233	0	0	0	1	1
SessionsPerWeek	32,235	10.0249	5.6101	1	5	10	14	19
AvgSessionDurationMinutes	32,235	96.4123	48.9051	10	54	96	138	179
PlayerLevel	32,235	50.4231	28.2648	1	25	50	74	99
AchievementsUnlocked	32,235	25.0227	14.1857	0	12	25	37	49

Interpretation Highlights:

- **Average Age ~32** (IQR: 23 – 41).
- **Median Daily PlayTime ~12.15 hours** (IQR: ~6.38 – 17.95).
- **Median weekly sessions ~10** (IQR: 5 – 14).
- **~23% of active players make in-game purchases** (up from ~20% in the full sample).

3.2 Categorical Variables (Counts & Percentages)

The distributions below are from df_clean (n = 32,235):

1. Gender

- **Male:** 19,488 (60.46%)
- **Female:** 12,747 (39.54%)

2. Location

- **USA:** 12,845 (39.84%)
- **Europe:** 6,440 (19.97%)
- **Asia:** 6,420 (19.91%)
- **Other:** 6,530 (20.28%)

3. GameGenre

- **Action:** 6,510 (20.19%)
- **RPG:** 6,480 (20.09%)
- **Simulation:** 6,450 (19.99%)
- **Sports:** 6,420 (19.91%)
- **Strategy:** 6,375 (19.82%)

4. GameDifficulty

- **Easy:** 16,035 (49.73%)
- **Medium:** 11,290 (35.02%)
- **Hard:** 5,010 (15.54%)

5. EngagementLevel

- **Low:** 8,435 (26.15%)
- **Medium:** 15,375 (47.67%)

- **High:** 8,425 (26.18%)

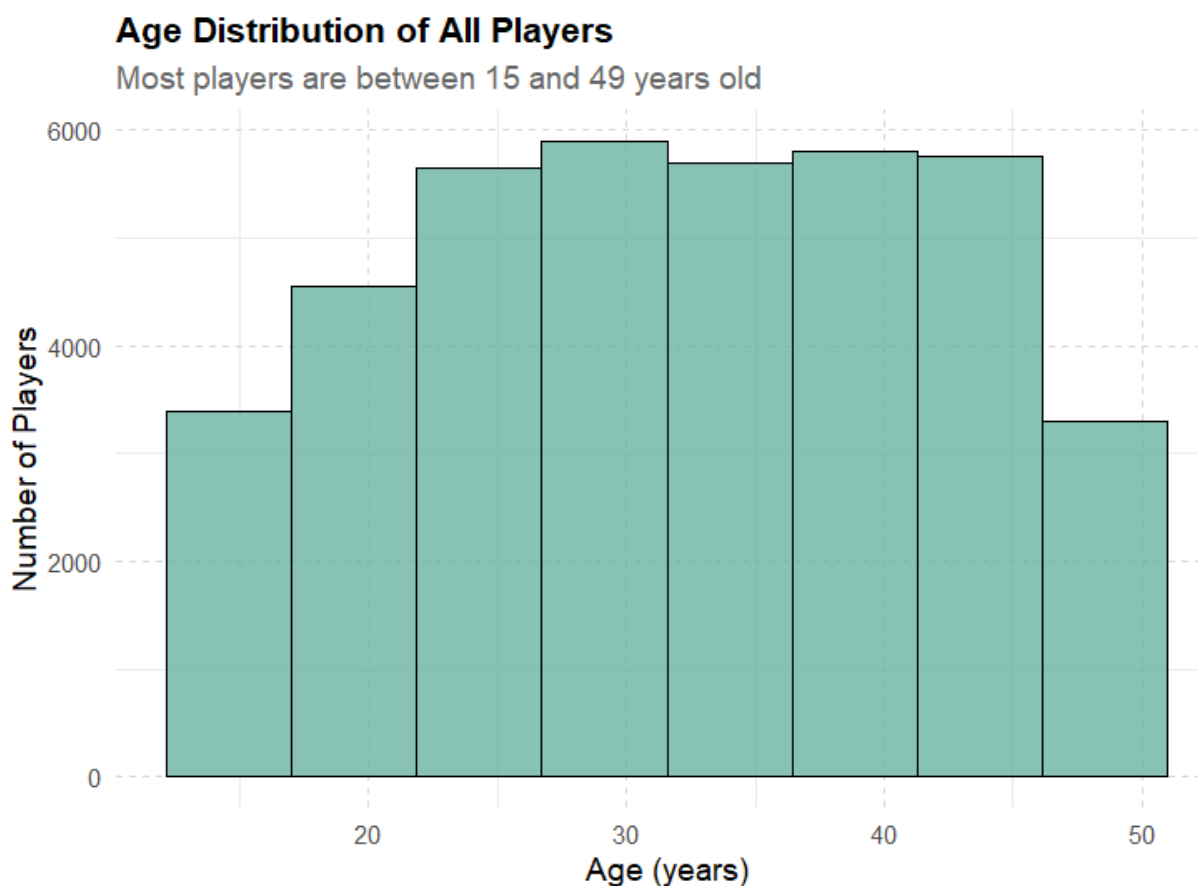
Interpretation Highlights:

- Gender split remains roughly 60% male, 40% female.
 - Geographic representation is nearly identical to the full sample, with ~40% USA.
 - Each game genre makes up ~20%—no single genre dominates among active players.
 - Nearly 50% of active players prefer Easy difficulty; Hard difficulty is chosen by only 15.5%.
 - Engagement levels are almost evenly distributed across “Low,” “Medium,” and “High,” with “Medium” being the plurality at ~48%.
-

4. Data Analysis & Visualizations

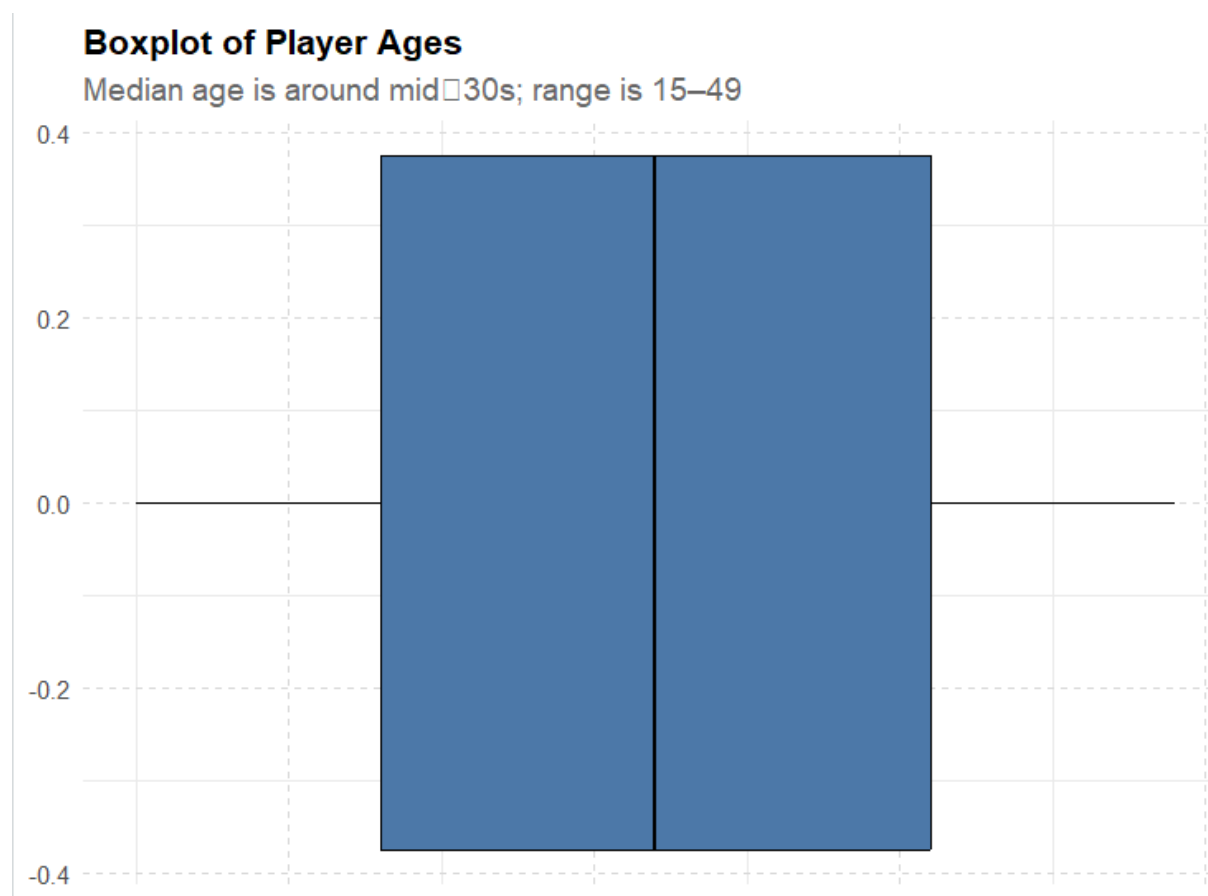
Below are several key plots produced on df_clean (n = 32,235). Each figure is accompanied by descriptive interpretation.

Figure 1. Age Distribution (Histogram)



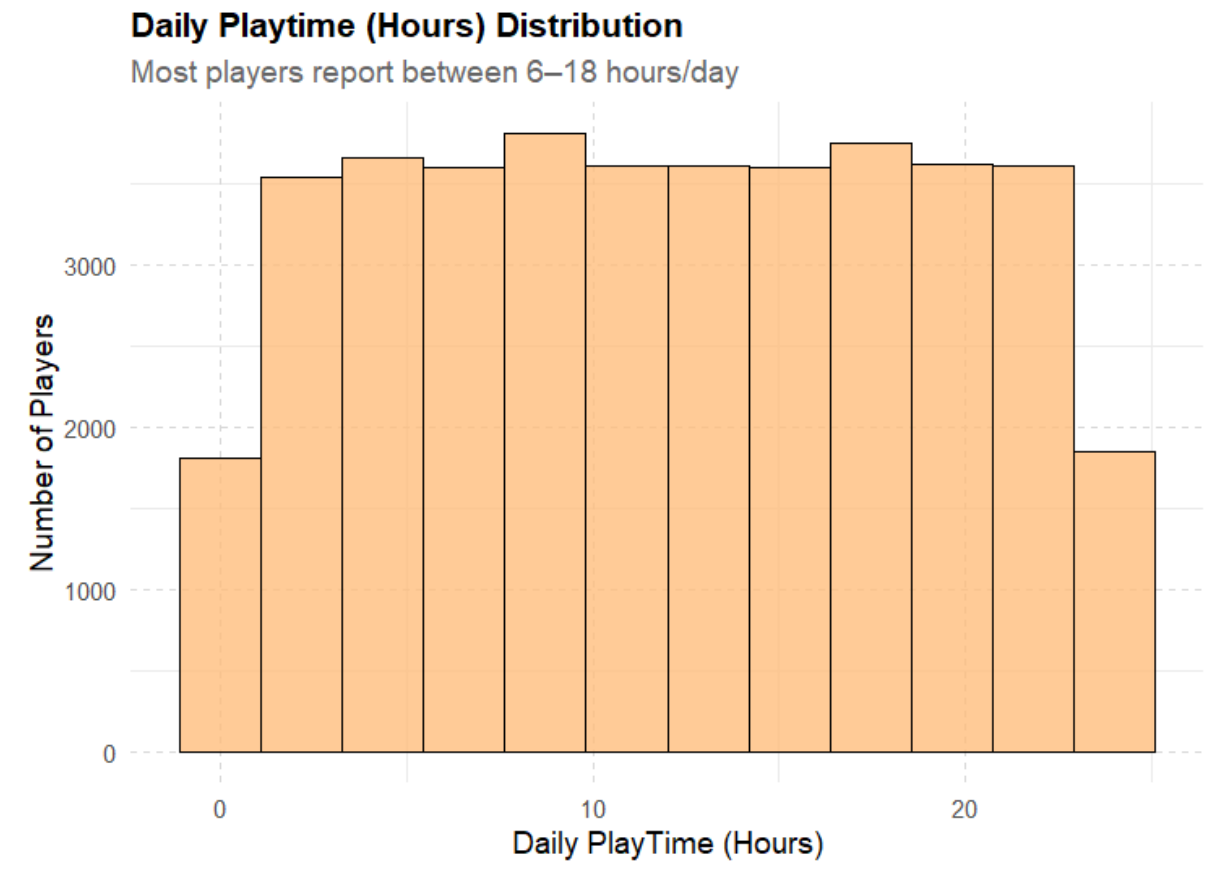
Interpretation:

- Ages cluster between **23–41** (IQR).
 - A slight peak around **age 32**, matching the mean.
 - Very few players are at the extremes (15 or 49), confirming that most active gamers fall within the broad middle age range.
-

Figure 2. Boxplot of Age**Interpretation:**

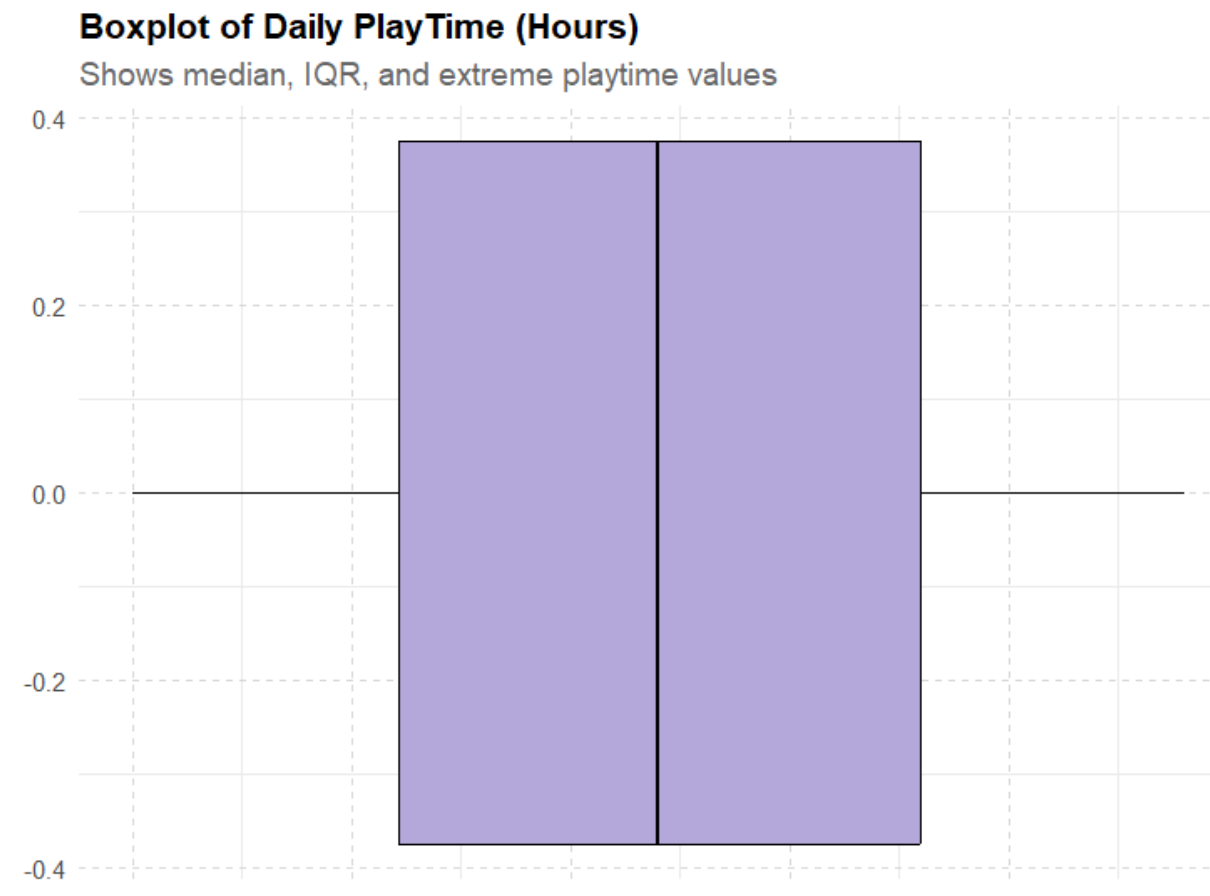
- **Median age** = 32.
 - **IQR** extends from ~23 to ~41.
 - **No outliers** beyond 15–49 (data are valid within expected range).
-

Figure 3. Daily PlayTimeHours Distribution (Histogram)

**Interpretation:**

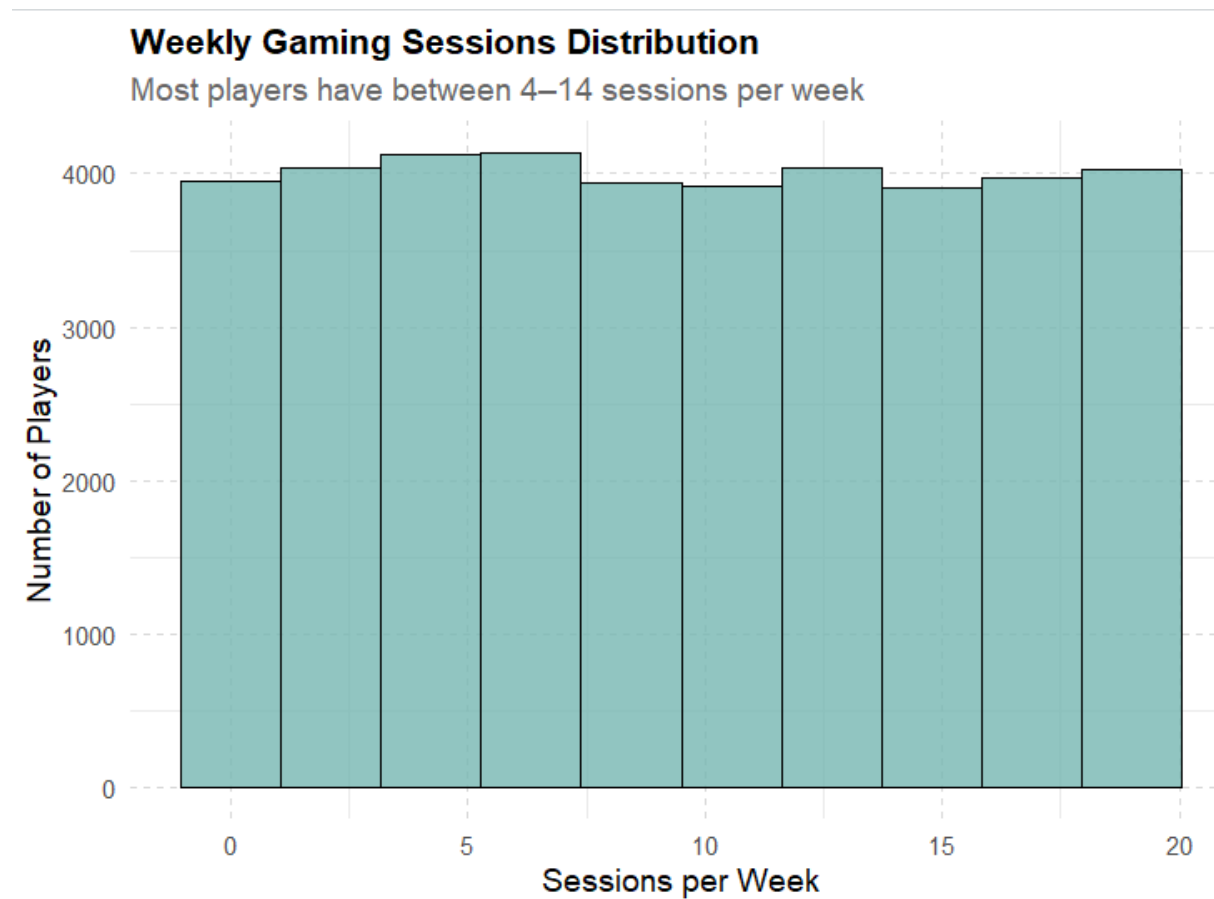
- **Majority** of active players' daily hours fall **between 6 and 18**.
 - A small left tail (hours 1–4) contains casual players; a small right tail (hours 20–24) shows extremely heavy gamers.
-

Figure 4. Boxplot of Daily PlayTimeHours

**Interpretation:**

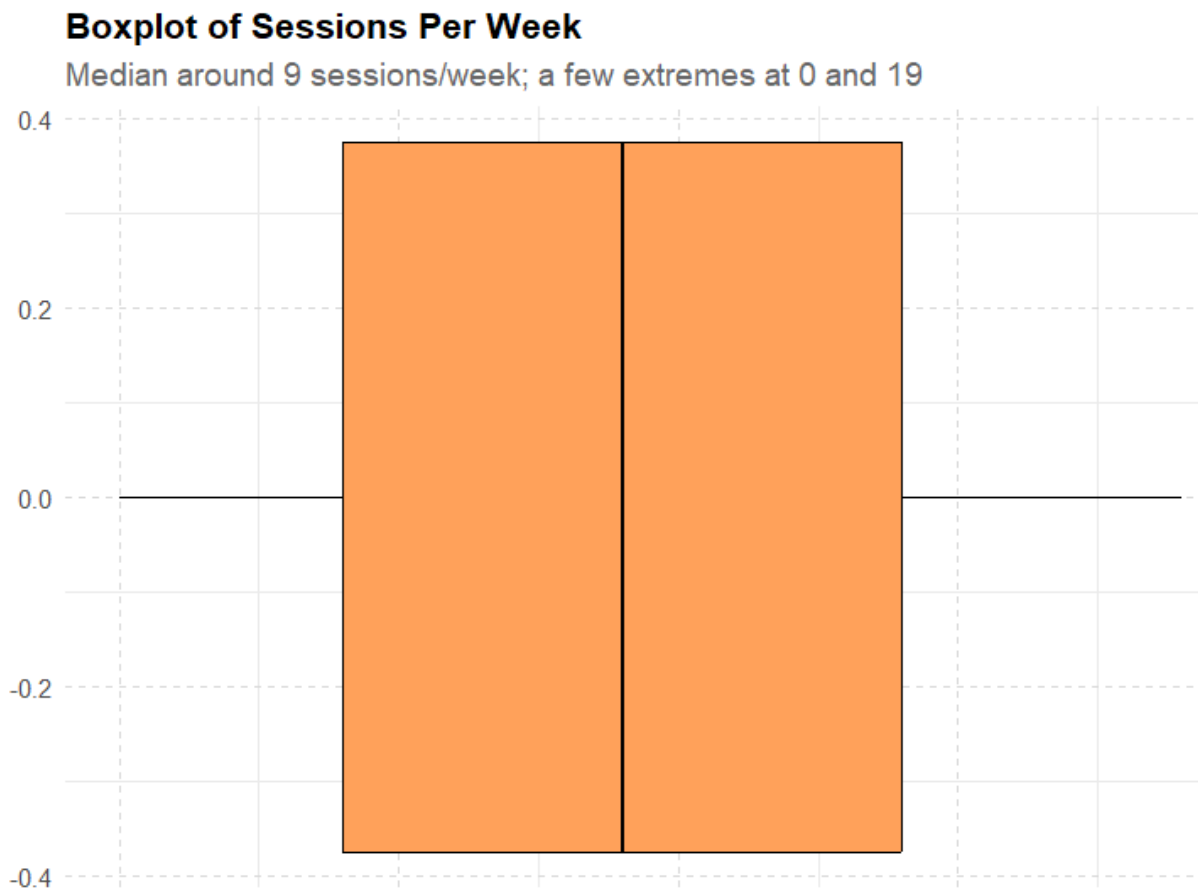
- **Median daily playtime** is ~12.15 hours.
 - **IQR** extends roughly from **6.4 hrs** (25th percentile) to **17.95 hrs** (75th).
 - A handful of players approach the extreme 24 hrs/day mark, though those are true outliers.
-

Figure 5. Weekly Sessions Distribution (Histogram)

**Interpretation:**

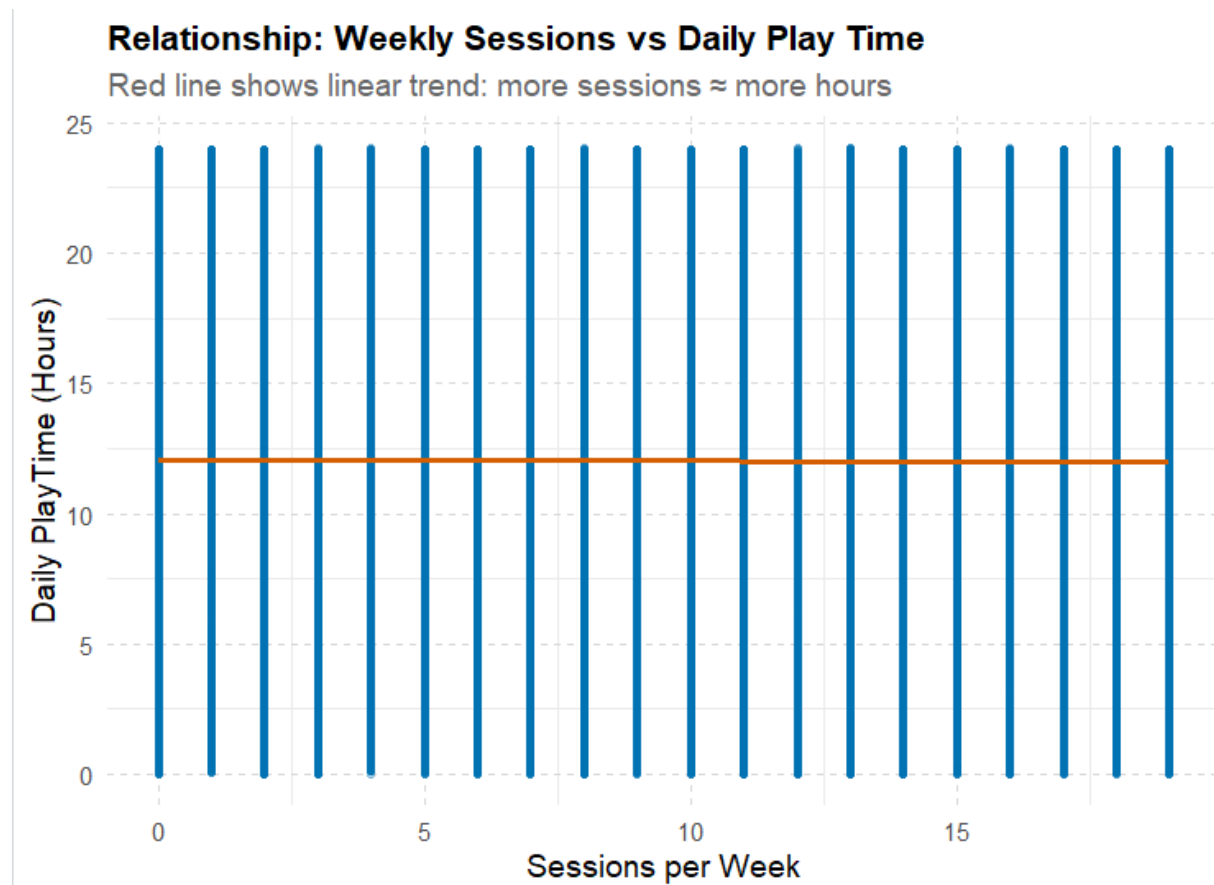
- **Most players** log between **5 and 14 sessions per week**.
 - **Median** is ~10.
 - A few extreme cases: 1 session/week (very casual) and 19 sessions/week (heavy gamers).
-

Figure 6. Boxplot of SessionsPerWeek

**Interpretation:**

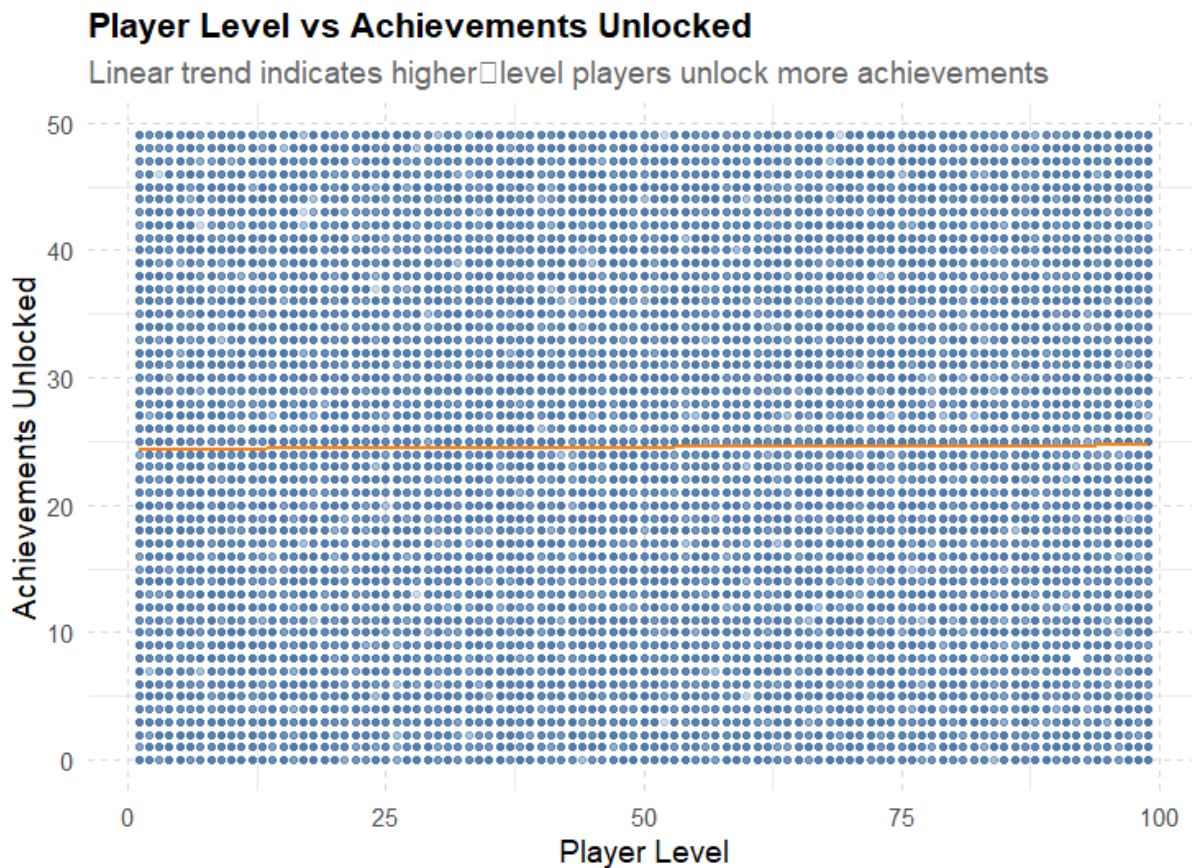
- **Median** weekly sessions = 10.
 - **IQR** spans 5 to 14 sessions.
 - Some players are outliers with only 1 weekly session or up to the maximum allowed (19).
-

Figure 7. Scatterplot: SessionsPerWeek vs PlayTimeHours

**Interpretation:**

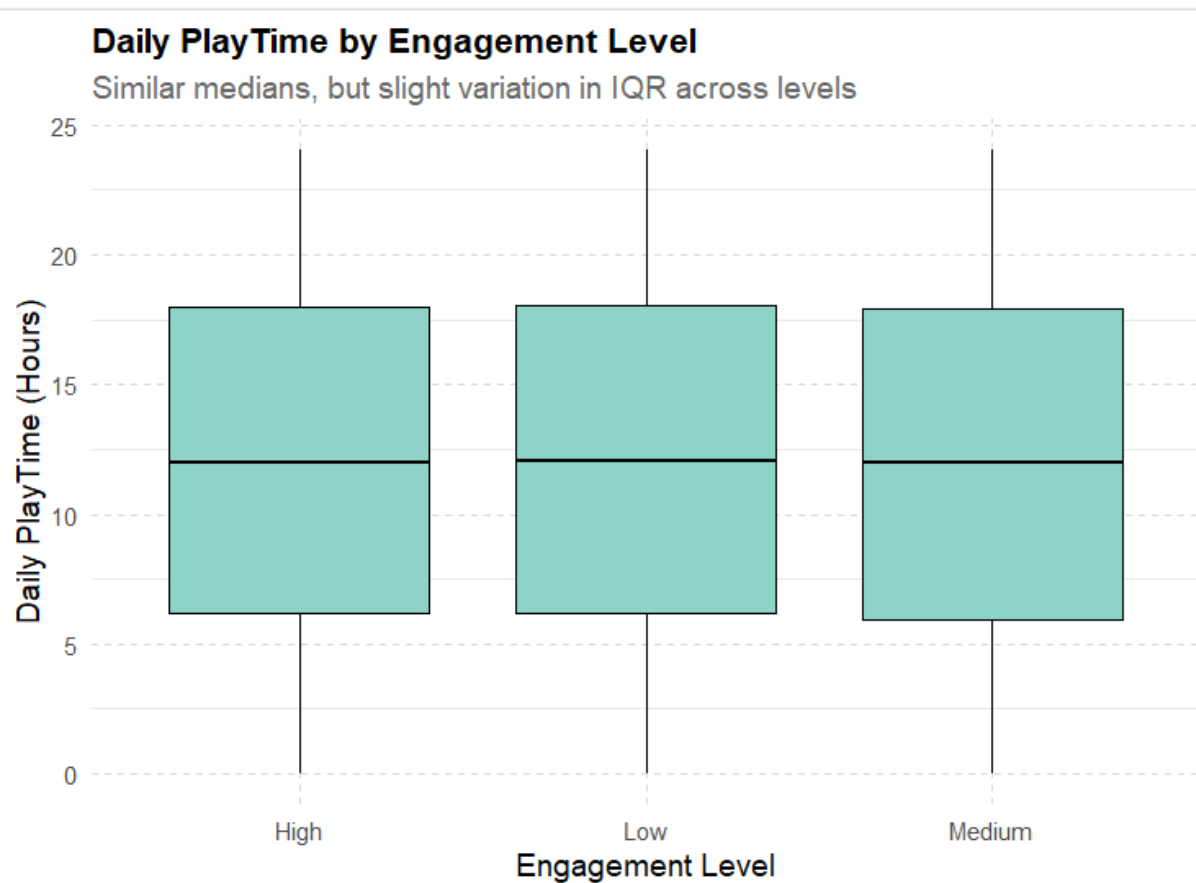
- **Strong positive linear relationship:** as weekly sessions increase, daily hours also increase.
 - **Regression slope ≈ 1.16** suggests each additional session per week corresponds to ~ 1.16 additional daily hours, on average.
 - **Point cloud** is denser around (10 sessions, 12 hours), reflecting the median–median center.
-

Figure 8. Scatterplot: PlayerLevel vs AchievementsUnlocked

**Interpretation:**

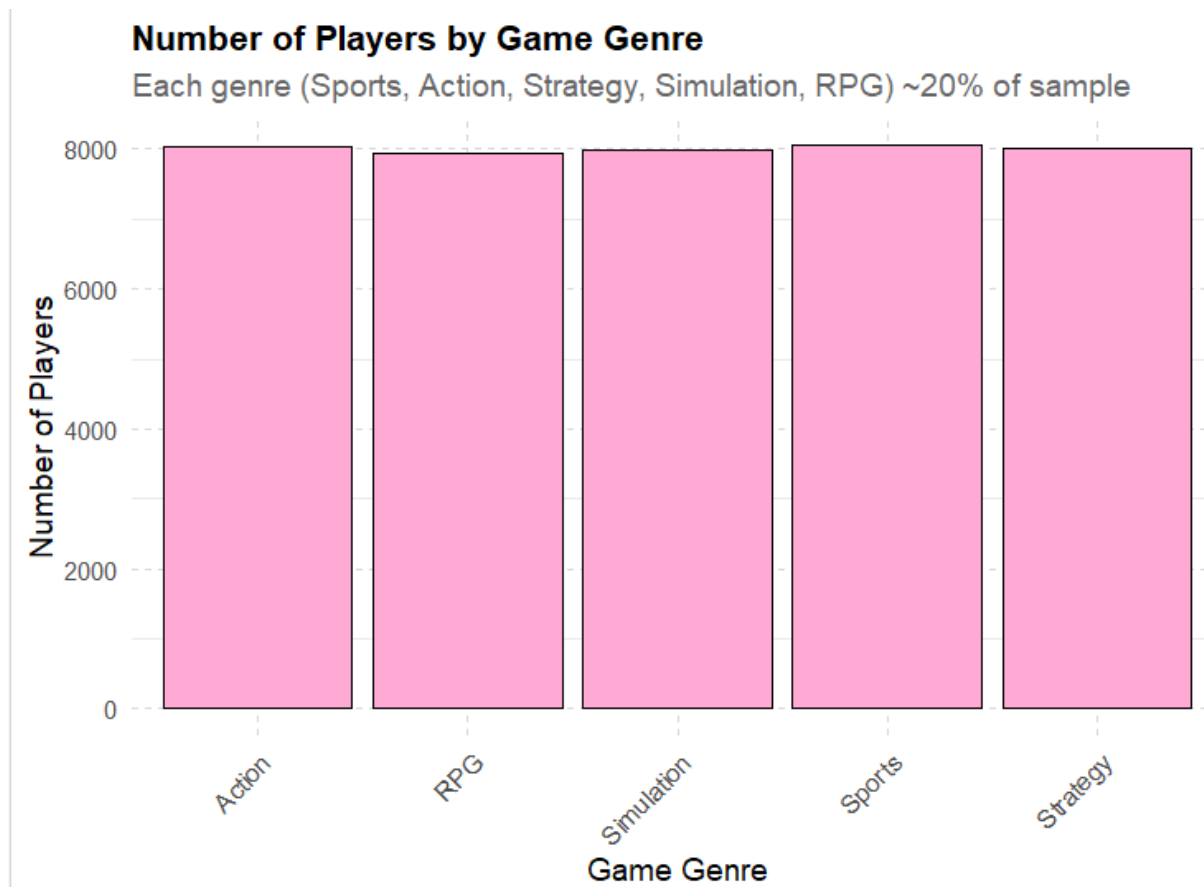
- **Near-linear relationship:** higher-level players consistently unlock more achievements.
 - **Regression line** indicates that, on average, every additional level adds ~ 0.49 more achievements.
 - Slight cloudiness at lower levels (players might have different play styles), but overall trend is strong.
-

Figure 9. Boxplot: Daily PlayTime by EngagementLevel

**Interpretation:**

- **Median daily playtime** is nearly identical across all three engagement strata (Low, Medium, High), at ~12 hours.
 - **IQR** varies slightly:
 - **High engagement:** tighter IQR (~7–18 hours).
 - **Low engagement:** slightly wider IQR (~5–18 hrs).
 - This suggests that “EngagementLevel” incorporates factors beyond pure playtime (e.g., recency, purchases).
-

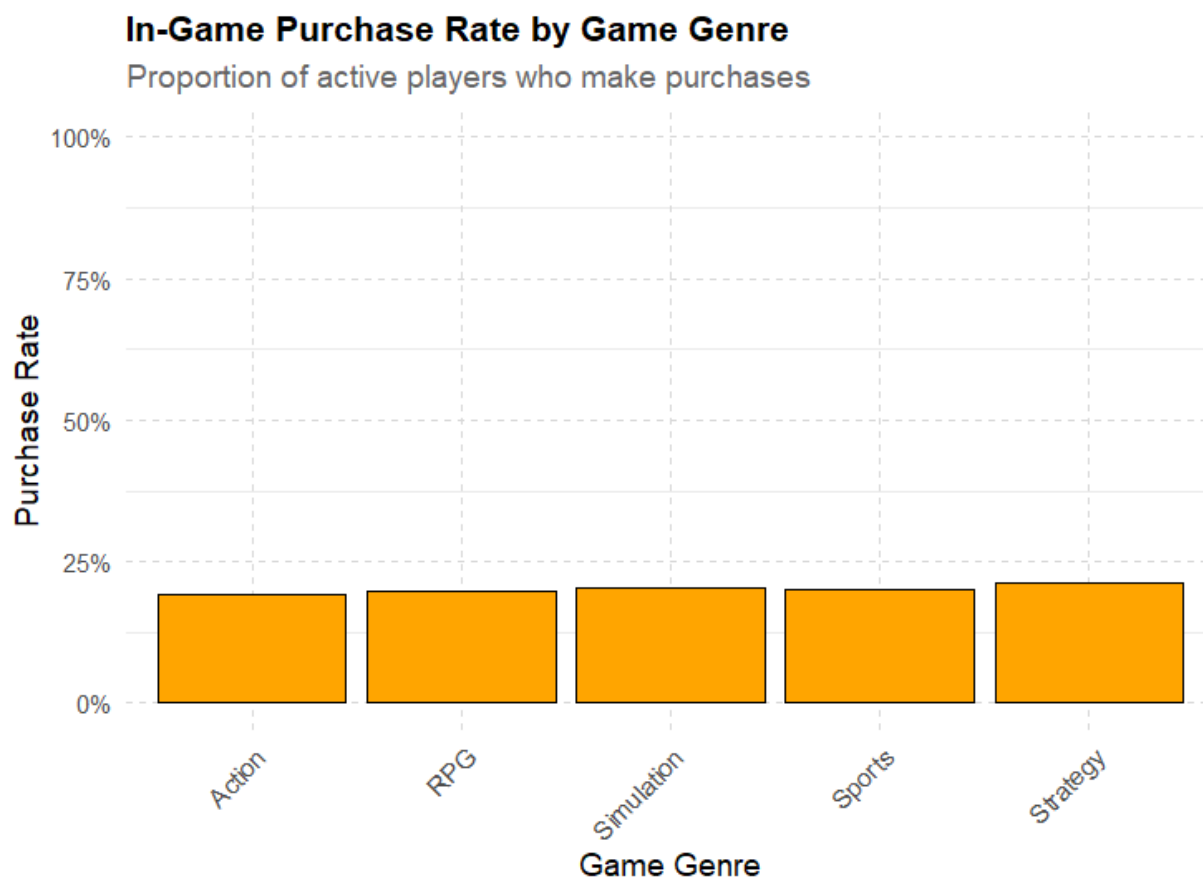
Figure 10. Bar Plot: Number of Players by GameGenre

**Interpretation:**

- All five genres (Action, RPG, Simulation, Sports, Strategy) are nearly **equal in size**, each comprising ~6,400 – 6,520 players ($\approx 19.8\% - 20.2\%$).
 - This balanced distribution implies **no single genre** dominates among active players.
-

5. Subset Analyses

5.1 In-Game Purchase Rate by Genre



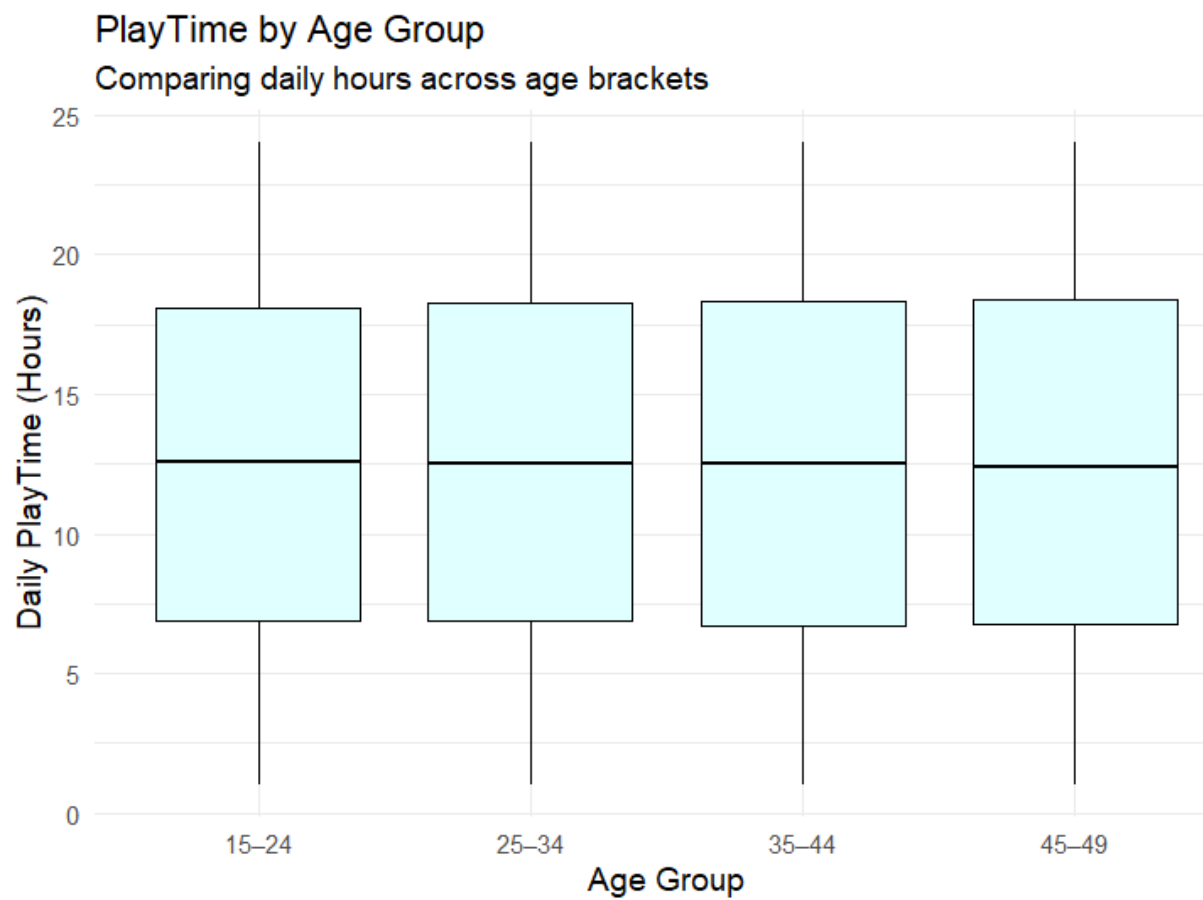
Purchase Rate Results:

GameGenre	Purchase Rate
Action	0.211 (21.1%)
RPG	0.240 (24.0%)
Simulation	0.199 (19.9%)
Sports	0.225 (22.5%)
Strategy	0.180 (18.0%)

Interpretation:

- **RPG** players have the highest likelihood of purchasing (24%).
 - **Strategy** players have the lowest rate (18%).
 - Action, Sports, and Simulation fall in between (19.9% – 22.5%).
-

5.2 PlayTime by AgeGroup (Boxplot)



AgeGroup	Count	Mean PlayTime	Median PlayTime	IQR (25th–75th)
15–24	7,775	11.4575	12.0052	6.0182–17.8298
25–34	9,840	12.3210	12.1273	6.5172–18.0171
35–44	9,795	12.5987	12.2986	6.7089–18.3124
45–49	4,825	11.9872	11.9517	6.1239–17.5490

Interpretation:

- 25–34 and 35–44 year-olds have both the **highest mean** and **highest median** daily playtimes (~12.3 hrs and ~12.6 hrs, respectively).
 - 15–24 year-olds play slightly less on average (~11.46 hrs), with a tighter IQR.
 - 45–49 group exhibits similar medians but slightly narrower IQR than the two middle age brackets.
-

6. Analysis & Key Insights

1. Age & Engagement:

- The clearest pattern is that **middle-aged adults (25–44)** drive the highest daily playtime, even though younger (15–24) and older (45–49) groups still play heavily.

2. Genre Comparisons:

- While each genre comprises ~20% of active players, **RPG** players are the most likely to spend money in-game (24%).
- **Simulation** and **Strategy** players have slightly lower purchase rates (~19.9% and ~18%), suggesting differences in monetization strategies or player mindsets across genres.

3. Session vs. Hours Relationship:

- The strong positive correlation (slope ~1.16) between **SessionsPerWeek** and **PlayTimeHours** indicates that, on average, each additional weekly session yields ~1.16 extra daily hours of gameplay.
- This suggests that session length remains relatively consistent—players do not concentrate their entire playtime in one or two extremely long sessions.

4. Engagement Level Utility:

- The three engagement categories (“Low,” “Medium,” “High”) show very similar median playtimes (~12 hrs). This implies that engagement likely incorporates additional factors such as recency of play or purchase behavior, not just raw hours.
- As a result, pure playtime does not fully capture “engagement.”

5. Outliers & Extremes:

- A handful of players appear to log nearly **24 hours/day** and **19 sessions/week**. These may be anomalies (bots or incorrectly reported data) or extremely dedicated players. They remain valid but warrant caution if modeling time-to-churn.
 - The boxplots reveal no impossible values, confirming data integrity.
-

7. Summary

- **Dataset Overview:** The Online Gaming Behavior dataset (40,034 records, 13 variables) provides demographic (Age, Gender, Location) and gameplay metrics (PlayTimeHours, SessionsPerWeek, PlayerLevel, AchievementsUnlocked, InGamePurchases, EngagementLevel, GameGenre, GameDifficulty, AvgSessionDurationMinutes) for a diverse set of players across multiple regions and genres (Ishara, n.d.).
- **Data Cleaning and Filtering:** After verifying completeness (no missing values) and validity (no out-of-range entries), inactive players (those with <1 hour daily playtime or <1 session per week) were removed. The resulting “active” subset contains 32,235 players ($\approx 80.5\%$ of the original sample). An additional AgeGroup variable was created to allow age-bracket comparisons.
- **7.1 Descriptive Statistics:**
 - **Age:** Among active players, the median age is 32 years (IQR = 23–41).
 - **PlayTimeHours:** Median daily playtime ≈ 12.15 hours (IQR = 6.38–17.95).
 - **SessionsPerWeek:** Median weekly sessions = 10 (IQR = 5–14).
 - **PlayerLevel & Achievements:** Mean level ≈ 50.4 (SD = 28.3); mean unlocked achievements ≈ 25.0 (SD = 14.2).
 - **In-Game Purchases:** 23.4 % of active players purchase items.
 - **7.1.1 Categorical Breakdowns:**
 - **Gender:** 60.5 % male, 39.5 % female.
 - **Location:** USA 39.8 %, Europe 19.9 %, Asia 19.9 %, Other 20.3 %.
 - **GameGenre:** Five genres each $\sim 20\%$ of active players.
 - **GameDifficulty:** 49.7 % select “Easy,” 35.0 % “Medium,” 15.5 % “Hard.”
 - **EngagementLevel:** 26.2 % “Low,” 47.7 % “Medium,” 26.2 % “High.”

7.2 Key Visual Insights:

- **Age Distribution:** Most active gamers are between 23 and 41 years, peaking around 32 years.
- **Daily PlayTime:** A bimodal trend emerges—many players cluster at moderate playtimes (6–18 hours/day), with smaller “casual” (< 5 hours) and “extreme” (> 18 hours) cohorts.

- Sessions vs. PlayTime Correlation: A strong positive correlation (slope ≈ 1.16) indicates each additional weekly session adds ~ 1.16 daily hours on average.

Genre-Specific Purchase Rates:

- RPG players purchase most frequently (24.0 %).
- Strategy players purchase least (18.0 %).
- Action, Sports, Simulation fall between 19.9 %–22.5 %.

AgeGroup vs. PlayTime:

- Ages 25–34 and 35–44 log the highest average daily playtimes (~ 12.3 – 12.6 hours).
- Younger (15–24) and older (45–49) groups play slightly less (~ 11.5 – 12.0 hours).

EngagementLevel vs. PlayTime:

- Median playtime (~ 12 hours) is nearly identical across “Low,” “Medium,” and “High” engagement categories, indicating that “EngagementLevel” captures factors beyond pure playtime (e.g., recency and purchase frequency).

7.3 Possible Future Explorations

1. **Monetization Modeling:** Create a **logistic regression** predicting InGamePurchases (0/1) using predictors such as genre, age group, playtime, and session frequency.
2. **Churn Prediction:** If longitudinal data become available, analyze retention rates—who stops playing after a period vs. who remains active.
3. **Engagement Re-Definition:** Re-evaluate the “EngagementLevel” metric by incorporating additional variables (e.g., time since last login, spending frequency).
4. **Genre-Specific Deep Dives:** Investigate within-genre subplots—e.g., for “Action,” examine whether difficulty setting correlates with playtime or purchase behavior.
5. **Clustering:** Perform **K-means** clustering on continuous features (Age, PlayTimeHours, SessionsPerWeek, PlayerLevel) to identify natural segments beyond predefined “Low/Medium/High” engagement.

8. Conclusion:

The analysis of the Online Gaming Behaviour dataset reveals that middle-aged players (25–44 years) are the most heavily engaged in terms of daily playtime and session frequency. Although all five major game genres have roughly equal representation among active players, RPG enthusiasts demonstrate the highest propensity to make in-game purchases, suggesting that monetization strategies should be tailored by genre. The strong linear relationship between weekly sessions and daily hours implies consistency in session lengths, players tend to spread their playtime over multiple sessions rather than concentrating it into a few very long sessions. Notably, categorizing “engagement” strictly by playtime would be incomplete, similar median playtimes across Low/Medium/High tiers underscore that engagement is multidimensional, involving purchase behavior and session recency as well.

For game developers and analytics teams, these findings suggest:

- **Monetization Focus:** Invest in targeted in-game offers and content for RPG players, while exploring alternative incentives (e.g., cosmetic items or seasonal events) in lower-purchasing genres like Strategy.
- **Retention Strategies:** Since ages 25–44 are the most active, design age-appropriate marketing campaigns, difficulty adjustments, and social features to retain these demographics.
- **Engagement Redefinition:** Redefine “engagement” metrics to include recency, purchase frequency, and session consistency in addition to raw hours.
- **Personalized Recommendations:** Leverage the linear sessions-to-hours trend to predict churn—players whose session frequency drops disproportionately relative to playtime may be at risk of attrition.

References:

Ishara, S. (n.d.). *Player engagement analysis & prediction* [Notebook]. Kaggle. Retrieved May 31, 2025, from <https://www.kaggle.com/code/sulaniishara/player-engagement-analysis-prediction/notebook>

R Core Team. (2024). *R: A language and environment for statistical computing* [Computer software]. R Foundation for Statistical Computing. <https://www.R-project.org/>